

Assignment for Week 5

Author: Your name here!

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Package Loading

```
library(dplyr, warn.conflicts = FALSE)
library(ggplot2)
library(haven)
```

Assignment

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- 1) Let E, F , and G be events. Find the expressions (using $\cap \cup$ and c) for the events that of E, F , and G
 - a) only F occurs,
 - b) both E and F but not G occur,
 - c) at least one event occurs,
 - d) at least two events occur,
 - e) all three events occur,
 - f) none occurs,
 - g) at most one occurs,
 - h) at most two occur.
- 2) 1.17 (p. 57)
- 3) 1.19 (p. 57)
- 4) 1.20 (a) (b) (p. 57)
- 5) 1.39 (p. 62)
- 6) 1.56 (a) (b) (p. 68)

- 7) Use R to simulate the following problems:
- (a) If three players are dealt five cards from a standard 52 card deck, how what is the probability that at least one person gets exactly two aces?
 - (b) Suppose that n letters are placed randomly in n envelopes, what is the probability that $n - 1$ letters will be placed in the correct envelope?
 - (c) Suppose that n letters are placed at random in n envelopes, if q_n is the probability that no letter is placed in the correct envelope, which of the following has the largest q_n ? $n = 10$, $n = 21$, $n = 53$, $n = 100$.
 - (d) If the n letters are placed in n envelopes, what is the expected number of letters that go to the correct destination if $n = 10$, $n = 20$, $n = 50$?

8)

```
PersonnelData <- data.frame(Representative = c(1:4),  
Sales = c(95,110,115,90), Territory = c(1:4))
```

Using `mapply()`, find the `class` of `PersonnelData`'s columns.

- 9) Using the ideas from the Foundations of Regression sheet, why are the following two regressions different? Note, -1 means leave out the intercept. Regressing on 1 would recover just the intercept.

```
x <- 1:10  
y <- 1/3*x + rnorm(10)  
lm( y ~ x - 1 )  
##  
## Call:  
## lm(formula = y ~ x - 1)  
##  
## Coefficients:  
##      x  
## 0.3234  
lm( x ~ y - 1 )  
##  
## Call:  
## lm(formula = x ~ y - 1)  
##  
## Coefficients:  
##      y  
## 2.891
```

- 10) Use only matrix multiplication `%*%` to get the coefficients from the regression from the previous problem.
- 11) Install the `fueleconomy` package. Using the `data(vehicles)` dataset, use `lm` to recover the average number of cylinders.
- 12) Use `lm` to recover the average number of cylinders by fuel type. Check your answer with `dplyr` `group_by` and `summarise`.
- 13) Using the `vehicles` dataset, estimate the relationship (slope coefficient) between the year of introduction and highway fuel economy for each class of vehicle, and plot the coefficients using `geom_points`, add `geom_errorbar` based on 2 times the standard error estimates of the coefficients.

The following code snippet gives you a starting point.

```
splitclass<-vehicles%>%split(.$class)  
splitclass %>% map( ~ lm(fuel~year, data=.x)) %>%map(summary)%>%map_dbl("r.squared")
```